

Lecture 04: Adversarial and Minimax Games

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Outline of MAAI

- 1 Basic Game Theory and Nash Equilibrium
- 2 Potential Games and Best Respond Dynamics
- 3 Repeated Games and the Theory of Cooperation
- 4 Adversarial and Minimax Games
- 5 General-Sum Games and L-H Algorithms
- 6 Single-agent Reinforcement Learning and MDP
- 7 Learning Stochastic Games
- 8 Non-regret Learning and Correlated Equilibrium
- 9 Counterfactual Regret Minimisation
- 10 Learning with a Population of Agents

- Subject to change

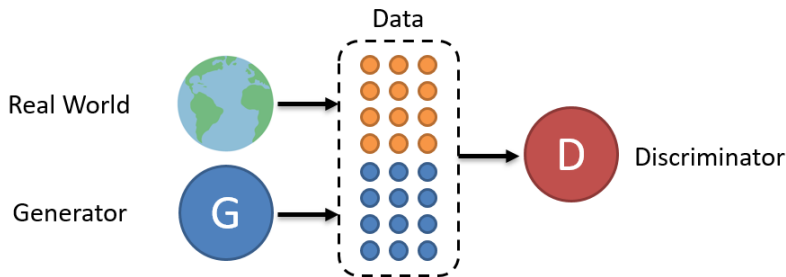
Outline

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 - Maxmin vs. Minimax
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 - LP connecting to Zero-Sum Games
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Generative Adversarial Nets (GAN)¹



Designed to learn $p_{G(Z)}$ to match a fixed data distribution p_{data} :

- Generator $G : Z \rightarrow X$ maps noise to data space (p_Z is given)
- Discriminator $D : X \rightarrow [0, 1]$ estimates probability of being real data
- Objective function (minimax game):

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{\text{data}}(x)} [\log D(x)] + \mathbb{E}_{z \sim p_Z(z)} [\log(1 - D(G(z)))]$$

¹Ian Goodfellow et al. "Generative adversarial nets". In: *Advances in neural information processing systems* 27 (2014).

GAN as a Normal Form Game

Normal Form Game Definition

A normal form game is a tuple $\Gamma = (N, (S_i)_{i \in N}, (u_i)_{i \in N})$, where:

- N : The set of players.
- S_i : The strategy space of player i .
- $u_i : \prod_{j \in N} S_j \rightarrow \mathbb{R}$: Player i 's payoff function.

For GANs, this becomes: $\Gamma_{GAN} = (\{G, D\}, (S_G, S_D), (u_G, u_D))$

- **Players:** $N = \{G, D\}$
- **Strategy Spaces:**
 - $S_G = \{f : Z \rightarrow X\}$ (Generator)
 - $S_D = \{h : X \rightarrow [0, 1]\}$ (Discriminator)
- **Payoff Functions:**
 - $u_D(s_G, s_D) = V(D, G)$
 - $u_G(s_G, s_D) = -V(D, G)$

Nash Equilibrium in GANs²

- Given a fixed G , one can find a the **optimal discriminator**:

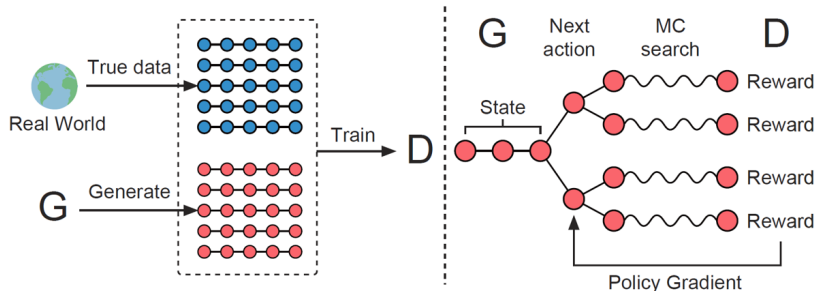
$$D_G^*(x) = \frac{p_{data}(x)}{p_{data}(x) + p_{G(Z)}(x)}$$

where $p_{G(Z)}$ is the generator's distribution.

- GAN **Nash Equilibrium** (when a generator G expressive enough to reproduce the distribution of observed samples):
 - Achieved when $p_{G(Z)} = p_{data}$
 - $D_G^*(x) = \frac{1}{2}$ for all x

²Farzan Farnia and Asuman Ozdaglar. "GANs may have no nash equilibria".
In: *arXiv preprint arXiv:2002.09124* (2020).

Sequential GAN (SeqGAN)³



- The **Generator** acts as a **reinforcement learning** policy that generates a sequence by deciding the next word to produce based on the previous ones.
- The **Discriminator** provides the **reward** (the probability of the sequence being real data) for the entire sequence.

³Lantao Yu et al. "SeqGAN: Sequence generative adversarial nets with policy gradient". In: *Proceedings of the AAAI conference on artificial intelligence*. Vol. 31. 1. 2017.

Self-play Large Language Models⁴

Chosen: ground truth completion



In Southampton, what is the most popular form of transportation for commuters?



I don't have access to current data. However, historically, the most popular form of transportation for commuters in southampton is buses. The city also has a good network of trains and taxis.



D_t Rejected: model generation @Iter 0

In Southampton, what is the most popular form of transportation for commuters?



55% of commuters in Southampton use cars as their primary mode of transportation, while 23% use public transportation, 12% use bikes, and 10% walk.

D_{t+1} Model generation @Iter 1



I do not have access to the latest updates and statistics of southampton. However, based on previous reports, the most popular form of transportation for commuters in southampton is the public transport network, which includes buses, trains, and ferries. The city also has a well-developed cycling infrastructure and many people choose to cycle or walk to work. Additionally, private cars and taxis are also commonly used for commuting.

- The Discriminator G_t aims to distinguish between the answer provided by the ground truth and that from a previous (weak) LLM model D_t .
- The Generator D_{t+1} is Large Language Models (LLM) which generate the answer.

⁴Zixiang Chen et al. "Self-play fine-tuning converts weak language models to strong language models". In: *arXiv preprint arXiv:2401.01335* (2024).

Another Example: Robust Learning⁵

Overview:

- In **model-based reinforcement learning**, one needs to learn a world model ϕ as well as a policy θ :

$$p_{\theta}^{\phi}(\tau) = \mu_0(s_0)\pi(a_0|s_0) \prod_{t=1}^{T-1} P_{\phi}(s_{t+1}|s_t, a_t)\pi_{\theta}(a_t|s_t).$$

- **Robustness** can be obtained by a **minimax game**:

$$\max_{\theta} \left[\min_{\phi} \mathbb{E}_{\tau \sim p_{\theta}^{\phi}(\tau)} [R_{\text{total}}(\tau)] \right],$$

where τ is the trajectory and R is the reward.

⁵Mohammed Amin Abdullah et al. "Wasserstein robust reinforcement learning". In: *arXiv preprint arXiv:1907.13196* (2019).

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Computing Nash Equilibria: A 2-person zero-sum games

	col 1	col 2
row 1	-2, +2	+3, -3
row 2	+3, -3	-4, +4

A 2-person zero-sum game

- Suppose we know that this game has no **pure-strategy Nash equilibria**.
- By Nash's theorem, it must have a **mixed-strategy Nash equilibrium**.
- With those conditions, how can we find it?

Computing Nash Equilibria: A 2-person zero-sum games

	col 1	col 2
row 1	-2, +2	+3, -3
row 2	+3, -3	-4, +4

A 2-person zero-sum game

- Let x_1 be the probability that player 1 (row player) plays action row 1, in the Nash equilibrium.
- So with probability $x_2 = 1 - x_1$, player 1 will play action row 2.
- Likewise, let y_1 be the probability that player 2 (col player) plays action col 1, in the Nash equilibrium.
- So with probability $y_2 = 1 - y_1$, player 2 will play action col 2.

Computing Nash Equilibria: A 2-person zero-sum games

	col 1	col 2
row 1	-2, +2	+3, -3
row 2	+3, -3	-4, +4

A 2-person zero-sum game

- Next, let's write down what we know about the outcomes in terms of x_1 and y_1 .

In equilibrium, player 1's expected payoff is:

$$x_1 y_1 (-2) + x_1 (1 - y_1) (+3) + (1 - x_1) y_1 (+3) + (1 - x_1) (1 - y_1) (-4)$$

In equilibrium, player 2's expected payoff is:

$$x_1 y_1 (+2) + x_1 (1 - y_1) (-3) + (1 - x_1) y_1 (-3) + (1 - x_1) (1 - y_1) (+4)$$

Computing Nash Equilibria: A 2-person zero-sum games

Observation:

	col 1	col 2
row 1	-2, +2	+3, -3
row 2	+3, -3	-4, +4

A 2-person zero-sum game

- If player 2 selects y_1 so that player 1 gets a higher utility by playing row 1 instead of row 2, then player 1 will always select row 1, i.e., $x_1 = 1$
- But that can't be an equilibrium! (Why not?)

In equilibrium, player 1's expected payoff is:

$$x_1 \underbrace{\left(y_1(-2) + (1 - y_1)(+3) \right)}_{\text{row 1's utility for row player}} + (1 - x_1) \underbrace{\left(y_1(+3) + (1 - y_1)(-4) \right)}_{\text{row 2's utility for row player}}$$

Computing Nash Equilibria: A 2-person zero-sum games

Observation:

	col 1	col 2
row 1	-2, +2	+3, -3
row 2	+3, -3	-4, +4

A 2-person zero-sum game

- Likewise, if player 1 selects x_1 so that player 2 gets a higher utility by playing col 2 instead of col 1, then player 2 will always select col 2.
- But that can't be an equilibrium, either! (as y_1 has to be larger than zero)

In equilibrium, player 2's expected payoff is:

$$y_1 \underbrace{\left(x_1(+2) + (1 - x_1)(-3) \right)}_{\text{col 1's utility for col player}} + (1 - y_1) \underbrace{\left(x_1(-3) + (1 - x_1)(+4) \right)}_{\text{col 2's utility for col player}}$$

Computing Nash Equilibria: A 2-person zero-sum games

	col 1	col 2
row 1	-2, +2	+3, -3
row 2	+3, -3	-4, +4

A 2-person zero-sum game

- In the latter case, the mixed equilibrium occurs when player 1 selects x_1 such that player 2's payoffs for choosing col 1 and col 2 are equal.

We thus have:

$$\underbrace{(x_1(+2) + (1 - x_1)(-3))}_{\text{col 1's utility for col player}} = \underbrace{(x_1(-3) + (1 - x_1)(+4))}_{\text{col 2's utility for col player}}$$
$$+5x_1 - 3 = -7x_1 + 4$$
$$12x_1 = 7 \longrightarrow x_1 = 7/12$$

For the former case, we have similarly:

$$\underbrace{(y_1(-2) + (1 - y_1)(+3))}_{\text{row 1's utility for row player}} = \underbrace{(y_1(+3) + (1 - y_1)(-4))}_{\text{row 2's utility for row player}} \longrightarrow y_1 = 7/12$$

Computing Nash Equilibria: A 2-person zero-sum games

	col 1	col 2
row 1	-2, +2	+3, -3
row 2	+3, -3	-4, +4

A 2-person zero-sum game

- In conclusion, now we know a mixed-strategy Nash equilibrium:

$$P_{\text{player 1}(\text{row 1})} = 7/12, \quad P_{\text{player 1}(\text{row 2})} = 5/12$$

$$P_{\text{player 2}(\text{col 1})} = 7/12, \quad P_{\text{player 2}(\text{col 2})} = 5/12$$

Quiz: A 2-person zero-sum games

	col 1	col 2
row 1	-2, +2	+3, -3
row 2	+3, -3	-4, +4

The value for Player 1 is the negative of the value for Player 2 because this is a zero-sum game.

The value of the game in a zero-sum game

is the expected payoff (from a maximiser view point) when both players use their optimal strategies. It represents the equilibrium outcome where neither player can improve their position by unilaterally changing their strategy.)

What is the value of this game ?

Answer: A 2-person zero-sum games

	col 1	col 2
row 1	-2, +2	+3, -3
row 2	+3, -3	-4, +4

You can get the value of the game three ways:

$$x_1 y_1 (-2) + x_1 (1 - y_1) (+3) + (1 - x_1) y_1 (+3) + (1 - x_1) (1 - y_1) (-4)$$

or:

$$y_1 (-2) + (1 - y_1) (+3)$$

or:

$$y_1 (+3) + (1 - y_1) (-4)$$

All these equal $1/12$.

Quiz: Computing an Equilibrium for Zero-Sum Games

	col 1	col 2
row 1	+5, -5	+2, -2
row 2	+3, -3	+6, -6

In equilibrium:

- 1 What is the probability that P_1 plays row 1?
- 2 What is the probability that P_2 plays col 1?
- 3 What is the value of the game for P_1 ?

Answer: Computing an Equilibrium for Zero-Sum Games

	col 1	col 2
row 1	+5, -5	+2, -2
row 2	+3, -3	+6, -6

Answers:

- Probability P_1 plays row 1: $1/2$.
- Probability P_2 plays col 1: $2/3$.
- Value of the game for P_1 : 4.

2-person Zero-Sum Games with More Actions

- When there are more actions available than 2 per person, the simple algorithm will no longer work.
- However, it is possible to compute Nash equilibria for zero-sum games in polynomial time using a technique called [Linear Programming](#).

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Zero-Sum Games

Consider the game of **Rock-Paper-Scissors**, where one player's gain is exactly equal to the other's loss:

	Rock	Paper	Scissors
Rock	0	-1	1
Paper	1	0	-1
Scissors	-1	1	0

The value V is called the **value (reward or payoff)** of the game. For example:

- The value of Rock-Paper-Scissors is 0.
- The best that Player 1 can hope to achieve, assuming Player 2 plays optimally (1/3 probability of each action), is a payoff of 0.

Minimax Theorem

Minimax Theorem (John von Neumann, 1928)⁶: For any finite two-player zero-sum game represented by an $m \times n$ payoff matrix A , let:

- X and Y be the mixed strategy spaces of the two players.
- The row player's worst-case payoff: $u(x) = \min_j xA_j$
- The column player's best-case payoff: $w(y) = \max_i A_i.y^t$, where t is transpose.
- Define: $u^* = \max_{x \in X} u(x)$, $w^* = \min_{y \in Y} w(y)$

Then, we have:

$$\max_{x \in X} \min_j xA_j = \min_{y \in Y} \max_i A_i.y^t = V. \quad (1)$$

⁶J v. Neumann. "Zur theorie der gesellschaftsspiele". In: *Mathematische annalen* 100.1 (1928), pp. 295–320.

Zero-Sum Games

- For every two-person, zero-sum game with finite pure strategies, there exists a mixed strategy for each player and a value V such that:

(Player 1 maximises V / Player 2 minimises V).

- Regardless of Player 2's strategy, the **best possible payoff** for Player 1 is V .
- Regardless of Player 1's strategy, the **best possible payoff** for Player 2 is $-V$.
- The existence is a **special case of Nash's theorem** and a predecessor to it.

Its elementary proof can be found in Appendix 3 (slide 76).

Zero-Sum Games (Continued)

- Player 1 can guarantee a payoff of **at least V** , and Player 2 can guarantee a payoff of **at least $-V$** (also stated as at most V).
- If both players play optimally, that's exactly what they will get:

$$V(\pi_1, \cdot) \leq V(\pi_1^*, \pi_2^*) \leq V(\cdot, \pi_2).$$

- It's called **"minimax"** because the players get this value by pursuing a strategy that minimises the maximum payoff of the other player.

Minimax Game Example

Let us look at penalty-kick game again:

kicker/goalkeeper	Defend Left	Defend Right
Left	0.58, -0.58	0.95, -0.95
Right	0.93, -0.93	0.70, -0.70

- Let's say the kicker announces ahead of time a strategy of $(1/2, 1/2)$.
- Presumably a "suboptimal" way to play the game, since it exposes the kicker's strategy to the goalkeeper who can now react optimally.
- Given the kicker's announced strategy, the goalkeeper will choose the strategy that maximises their payoff (value).
- In this case, what will the goalkeeper choose?
 - Defend left: $-0.58 \times \frac{1}{2} - 0.93 \times \frac{1}{2} = -0.755 \checkmark$
 - Defend right: $-0.95 \times \frac{1}{2} - 0.70 \times \frac{1}{2} = -0.825$

Minimax Game Example (Continued)

Penalty-kick game:

kicker/goalkeeper	Defend Left	Defend Right
Left	0.58, -0.58	0.95, -0.95
Right	0.93, -0.93	0.70, -0.70

- More generally, if the kicker announces strategy (x_1, x_2) , then the goalkeeper has the following expected payoffs:

$$V(\text{"Defend left"}) = -0.58x_1 - 0.93x_2,$$

$$V(\text{"Defend right"}) = -0.95x_1 - 0.70x_2.$$

- Thus the goalkeeper's optimal reaction to the kicker's announced strategy is:

$$\max(-0.58x_1 - 0.93x_2, -0.95x_1 - 0.70x_2).$$

- Due to a zero-sum game, the resulting goalkeeper's payoff is:

$$\min(0.58x_1 + 0.93x_2, 0.95x_1 + 0.70x_2).$$

Minimax Game Example (Continued)

- So if the kicker is forced to announce their strategy in advance, they will choose:

$$(x_1, x_2) = \arg \max_{x_1, x_2} \min(0.58x_1 + 0.93x_2, 0.95x_1 + 0.70x_2).$$

- In other words, the kicker's **optimal announced strategy** (x_1, x_2) can be computed via the following linear program:

maximise z

subject to $0.58x_1 + 0.93x_2 \geq z$,

$0.95x_1 + 0.70x_2 \geq z$,

$x_1 + x_2 = 1, \quad x_1, x_2 \geq 0$.

- The optimal z (**lower bound**) will be kicker's payoff after goalkeeper's optimal response to (x_1, x_2) . Solving this yields:

$$x_1 = 0.38, x_2 = 0.62, z = 0.8.$$

Minimax Game Example (Continued)

- Conversely, if the goalkeeper were forced to announce their strategy in advance, they would aim to solve the following:

maximise w ,

subject to $-0.58y_1 - 0.95y_2 \geq w$,

$-0.93y_1 - 0.70y_2 \geq w$,

$y_1 + y_2 = 1, \quad y_1, y_2 \geq 0$.

- This yields the following solution:

$$y_1 = 0.42, y_2 = 0.58, w = -0.80.$$

Minimax Game Example (Continued)

An interesting observation: If the kicker plays $x = \langle 0.38, 0.62 \rangle$ and the goalkeeper plays $y = \langle 0.42, 0.58 \rangle$, then:

- The payoff of the kicker is 0.80 and the payoff of the goalkeeper is -0.80 . The sum of their payoffs is zero.
- Given strategy $x = \langle 0.38, 0.62 \rangle$, the goalkeeper cannot obtain a payoff **greater than -0.80** .
 - since the kicker gets a payoff of at least 0.80 irrespective of the goalkeeper's strategy (by the first LP, kicker's payoff $\geq z$) and the game is zero-sum) $\rightarrow$ goalkeeper's payoff at most -0.80 .
- Similarly, given $y = \langle 0.42, 0.58 \rangle$, the kicker cannot obtain a payoff **more than 0.80**.
 - since goalkeeper gets payoff at least -0.80 irrespective of the kicker's strategy (the second LP, goalkeeper's payoff $\geq w$). $\rightarrow$ kicker's payoff at most 0.80.

Minimax Game Example (Continued)

	Defend Left	Defend Right
Left	0.58, -0.58	0.95, -0.95
Right	0.93, -0.93	0.70, -0.70

- Thus, the pair of strategies $x = \langle 0.38, 0.62 \rangle$ & $y = \langle 0.42, 0.58 \rangle$ is a **Nash equilibrium** (best response to each other).
- This was somewhat unexpected since the two players formulated the two LPs **independently** without any care about what their opponent might do.
 - In particular, it seemed a priori **suboptimal** for a player to announce their strategy and let the other player respond.
 - Also, no reason to expect that the suboptimal strategies for the two players are in fact best responses to each other.
- As we show next, this coincidence happens for a deep reason: it is true in all two-player zero-sum games, as a result of the **strong LP duality**.

Duality in Linear Programming

- One of the most important discoveries in the early development of linear program was the concept of **duality**.
- Every linear programming problem is associated with another linear programming problem called the **dual**.
- The relationships between the dual problem and the original problem (called the **primal**) prove to be extremely useful in a variety of ways.

We refer to section 2 in⁷ for an intuitive explanation of duality in linear programming.

⁷Princeton University COS 521 Staff. *Lecture Notes 7-8: Minimax Theorem and Zero-Sum Games*. <https://www.cs.princeton.edu/courses/archive/spring13/cos521/notes/lecture7-8.pdf>. Accessed: January 12, 2025. 2013.

Duality in Linear Programming (Continued)

Primal Problem:

Maximise $z = \mathbf{c}^T \mathbf{x}$,

subject to $\mathbf{Ax} \leq \mathbf{b}$,

$\mathbf{x} \geq 0$.

Dual Problem:

Minimise $w = \mathbf{y}^T \mathbf{b}$,

subject to $\mathbf{y}^T \mathbf{A} \geq \mathbf{c}^T$,

$\mathbf{y} \geq 0$.

- Note that $\mathbf{c}, \mathbf{x}$ are $n \times 1$ column vectors and $\mathbf{b}, \mathbf{y}$ are $m \times 1$ column vectors and $\mathbf{A}$ is $m \times n$ matrix.
- The dual problem uses exactly the same parameters as the primal problem, but in different locations.

Primal-Dual Table for Linear Programming

		Primal Problem						
		Coefficient of:				Right Side		
							x_1	
Dual Problem	Coefficient of:	y_1	a_{11}	a_{12}	$\dots$	a_{1n}	$\leq b_1$	Coefficients for Objective Function (Minimize)
		y_2	a_{21}	a_{22}	$\dots$	a_{2n}	$\leq b_2$	
		$\vdots$					$\vdots$	
		y_m	a_{m1}	a_{m2}	$\dots$	a_{mn}	$\leq b_m$	
		Right Side	VI	VI	$\dots$	VI		
		c_1	c_2	$\dots$	c_n			
		Coefficients for Objective Function (Maximize)						

Example

Primal Problem:

$$\text{Maximise } Z = \begin{bmatrix} 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix},$$

$$\text{subject to } \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \leq \begin{bmatrix} 4 \\ 12 \\ 18 \end{bmatrix},$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \geq \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Dual Problem:

$$\text{Minimise } W = \begin{bmatrix} y_1 & y_2 & y_3 \end{bmatrix} \begin{bmatrix} 4 \\ 12 \\ 18 \end{bmatrix},$$

$$\text{subject to } \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 2 \end{bmatrix}^T \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \geq \begin{bmatrix} 3 \\ 5 \end{bmatrix},$$

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \geq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Weak Duality and Strong Duality Theorems of LP

- The following relation is always maintained:

$$\mathbf{y}^T \mathbf{Ax} \leq \mathbf{y}^T \mathbf{b} \quad (\text{from Primal: } \mathbf{Ax} \leq \mathbf{b}), \quad (2)$$

$$\mathbf{y}^T \mathbf{Ax} \geq \mathbf{c}^T \mathbf{x} \quad (\text{from Dual: } \mathbf{y}^T \mathbf{A} \geq \mathbf{c}). \quad (3)$$

- From (1) and (2), we have:

$$\mathbf{c}^T \mathbf{x} \leq \mathbf{y}^T \mathbf{Ax} \leq \mathbf{y}^T \mathbf{b} \quad (\text{Weak Duality}). \quad (4)$$

- At optimality:

$$\mathbf{c}^T \mathbf{x}^* = (\mathbf{y}^*)^T \mathbf{Ax}^* = (\mathbf{y}^*)^T \mathbf{b} \quad (\text{Strong Duality}). \quad (5)$$

* denotes optimal value.

Complementary Slackness Conditions

- Derived from LP Strong Duality Theorem (5):

$$(\mathbf{c}^T - (\mathbf{y}^*)^T \mathbf{A})\mathbf{x}^* = 0, \quad (6)$$

$$(\mathbf{y}^*)^T (\mathbf{b} - \mathbf{A}\mathbf{x}^*) = 0. \quad (7)$$

- If $\mathbf{x}_j^* > 0$, then $\mathbf{y}^* \mathbf{a}_j = \mathbf{c}_j$, otherwise $\mathbf{x}_j^* = 0$.
- If $\mathbf{y}_i^* > 0$, then $\mathbf{a}_i^T \mathbf{x}^* = \mathbf{b}_i$, otherwise $\mathbf{y}_i^* = 0$.

The Dual of a Dual

- Any pair of primal and dual problems can be converted to each other.
- The dual of a dual problem always is the primal problem.

Back to General Two-Player Zero-Sum Games

- Recall that a two-player game is zero-sum if and only if the payoff matrices (R, C) satisfy $R + C = 0$.
- The LP for the row player if he is forced to announce his strategy in advance:

$$\begin{aligned} \max z \quad \text{s.t.} \quad & x^T R \geq z \mathbf{1}^T, \\ & x^T \mathbf{1} = 1, \quad \forall i, \quad x_i \geq 0. \end{aligned} \quad (\text{LP1}) \quad (8)$$

- The maximum of z is the same as:

$$\max_x \min_y x^T R y.$$

The Dual of LP1

- Consider its dual (for detailed conversion, we refer to Appendix 1 slide 69 or slide 44) as:

$$\begin{aligned} \min z' \quad \text{s.t.} \quad & -y^T R^T + z' \mathbf{1}^T \geq 0, \\ & y^T \mathbf{1} = 1, \quad \forall j, \quad y_j \geq 0. \end{aligned} \quad (\text{LP2}) \quad (9)$$

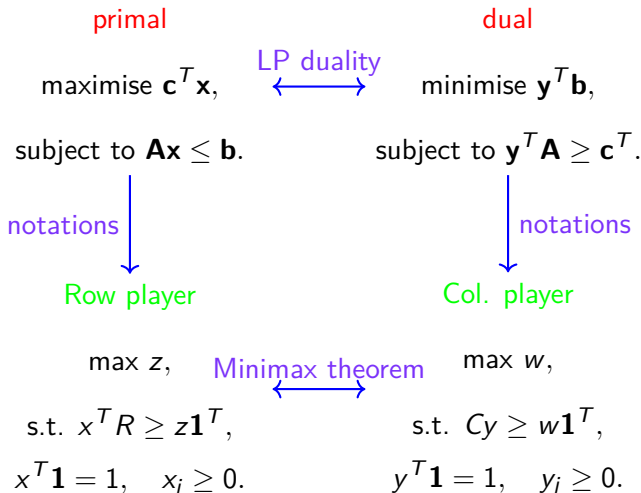
- Let us define $w = -z'$ and use the fact that $C = -R$ to change LP(2) into LP(3) below, whose value has flipped sign:

$$\begin{aligned} \max w \quad \text{s.t.} \quad & Cy \geq w \mathbf{1}^T, \\ & y^T \mathbf{1} = 1, \quad \forall j, \quad y_j \geq 0. \end{aligned} \quad (\text{LP3}) \quad (10)$$

- Recall:** LP(3) is the LP that the column player would solve if they were forced to announce their strategy in advance:

$$\max_y \min_x x^T Cy, \quad \text{or} \quad \min_y \max_x x^T Ry$$

Overview: Conversions between LPs and Zero-Sum Games



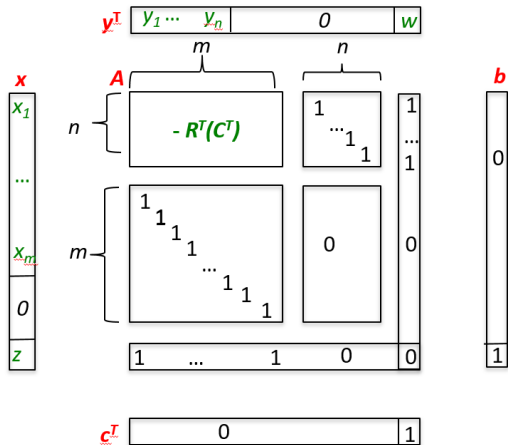
Conversion between Primal and Dual Problems

Primal Problem: Max $\mathbf{c}^T \mathbf{x}$,

Dual Problem: Min $\mathbf{y}^T \mathbf{b}$,

subject to $\mathbf{Ax} \leq \mathbf{b}$, ($\mathbf{x} \geq 0$).

subject to $\mathbf{y}^T \mathbf{A} \geq \mathbf{c}^T$, ($\mathbf{y} \geq 0$).



Red symbols are standard LP notations, whereas green ones are specific for zero-sum games. $\mathbf{x}$ concatenates mixed strategy $\mathbf{x} = \{x_1, \dots, x_m\}$ and value of the game z .

The difference compared to Appendix 1 (slide 69) is that this formulation directly incorporates the constraints $\mathbf{x} \geq 0, \mathbf{y} \geq 0$.

Conversion between Row Player and Column Player

Row Player (Primal):

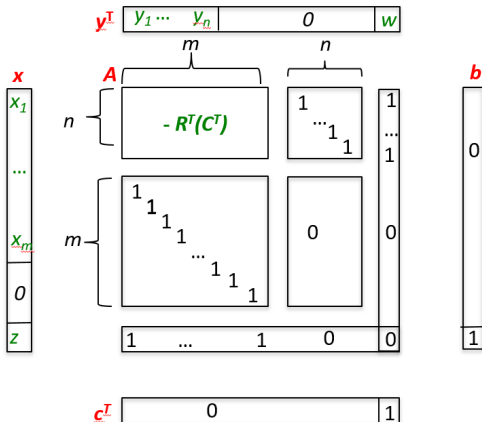
$$\max z \quad \text{s.t.} \quad x^T R \geq z \mathbf{1}^T,$$

$$x^T \mathbf{1} = 1, \quad \forall i, \quad x_i \geq 0.$$

Column Player (Dual):

$$\max w \quad \text{s.t.} \quad Cy \geq w \mathbf{1}^T,$$

$$y^T \mathbf{1} = 1, \quad \forall j, \quad y_j \geq 0.$$



Optimal Value

- Since LP(2) is the dual of LP(1), the LP strong duality theorem (Eq. 5) implies the following:
 - If (x^*, z^*) is optimal for LP(1) and (y^*, z'^*) is optimal for LP(2), then $z^* = z'^*$.
 - On the other hand, given our construction of LP(3), it follows that, if (y^*, z'^*) is optimal for LP(2), then (y^*, w^*) is optimal for LP(3), and vice versa (since $w^* = -z'^*$ by construction).
- Thus, if (x^*, z^*) is optimal for LP(1) and (y^*, w^*) is optimal for LP(3), then $z^* = w^*$.

Solution of LP is Nash Equilibrium

Theorem LP $\rightarrow$ NE:

If (x^*, z^*) is optimal for LP(1), and (y^*, w^*) is optimal for LP(3), then (x^*, y^*) is a Nash equilibrium of (R, C) .

Also, the payoffs of the row/column player in this Nash equilibrium are z^* and $w^* = -z^*$ respectively.

Proof Goal: From strong duality of LP,

$$z^* = -w^* \quad (\text{optimal values}) \quad \Rightarrow \quad (x^*, y^*) \text{ is NE.}$$

Proof of the Theorem (Part 1)

Step 1: Feasibility

- Since (x, z) is feasible for LP(1):

$$x^T R \geq z\mathbf{1}, \quad \Rightarrow \quad \forall y \quad x^T R y \geq z. \quad (11)$$

- Since (y, w) is feasible for LP(3):

$$C y \geq w\mathbf{1}, \quad \Rightarrow \quad \forall x \quad x^T C y \geq w. \quad (12)$$

$$\Rightarrow \forall x \quad x^T R y \leq -w. \quad (13)$$

Step 2: Bound Payoffs

- Inequality (13): $x^T R y \leq -w$ shows that if the column player plays y^* , the row player's payoff is at most z^* because $-w^* = z^*$ from strong duality:

$$\forall x \quad x^T R y^* \leq z^*. \quad (14)$$

Proof of the Theorem (Part 2)

Step 3: Best Responses

- Inequality (11): $\forall y, x^T R y \geq z$ gives $(x^*)^T R y^* \geq z^*$.
- But we know that $\forall x, x^T R y^* \leq z^*$ so x^* achieved the best response: $(x^*)^T R y^* = z^*$.
- Similarly, y^* is a best response for the column player, achieving $w^* = -z^*$.

Conclusion: (x^*, y^*) is a Nash equilibrium, where the row player's payoff is z^* , and the column player's payoff is $w^* = -z^*$.

Can Every Nash Equilibrium Be Found by LP?

Theorem NE $\rightarrow$ LP:

If (x^*, y^*) is a Nash equilibrium of (R, C) , then:

- $(x^*)^T R y^*$ is an optimal solution of LP(1),
- $-(x^*)^T C y^*$ is an optimal solution of LP(2).

Proof of the Theorem

- ① Let $z^* = (x^*)^\top Ry^* = -(x^*)^\top Cy^*$.
- ② Since (x^*, y^*) is a Nash equilibrium, we have:

$$(x^*)^\top Cy^* \geq (x^*)^\top C\mathbf{e}_j, \quad \forall j,$$

where $\mathbf{e}_j$ represents the unit vector for column j .

- ③ Rearrange terms gives: $-(x^*)^\top Ry^* \geq -(x^*)^\top R\mathbf{e}_j, \quad \forall j$.
- ④ Replacing $z^* = (x^*)^\top Ry^*$ gives: $(x^*)^\top R \geq z^*\mathbf{1}$.
- ⑤ Therefore, (x^*, z^*) is a feasible solution for LP(1).
- ⑥ By the same token, (y^*, z^*) is a feasible solution for LP(2).
- ⑦ LP(2) is the dual of LP(1). By **weak duality**, the value achieved by any feasible solution of LP(2) upper bounds the value achieved by any feasible solution of LP(1).
- ⑧ In this case, the values are **equal**. Thus (x^*, z^*) and (y^*, z^*) must be optimal solutions for LP(1) and LP(2), respectively.

Example: LP for Zero-Sum Games

Assume we have a 2×2 zero-sum matrix game given as:

$$R_1 = \begin{bmatrix} r_{11} & r_{12} \\ r_{21} & r_{22} \end{bmatrix}, \quad R_2 = -R_1$$

where R_1 is player 1's reward matrix (R_2 is player 2's).

p_j ($j = 1, 2$) is player 1's mixed strategy and q_j is player 2's.

The LP for Player 1: Find (p_1, p_2) to maximise V_1

$$\begin{aligned} \text{Subject to:} \quad & r_{11}p_1 + r_{21}p_2 \geq V_1, \quad r_{12}p_1 + r_{22}p_2 \geq V_1, \\ & p_1 + p_2 = 1, \quad p_1, p_2 \geq 0. \end{aligned}$$

The LP for Player 2: Find (q_1, q_2) to maximise V_2

$$\begin{aligned} \text{Subject to:} \quad & -r_{11}q_1 - r_{12}q_2 \leq V_2, \quad -r_{21}q_1 - r_{22}q_2 \leq V_2 \\ & q_1 + q_2 = 1, \quad q_1, q_2 \geq 0. \end{aligned}$$

LP is typically solved by the **simplex method**.

LP for Matching Pennies Game

Assume we have a 2×2 zero-sum matrix game given as:

$$R_1 = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}.$$

LP for Player 1: Since $p_2 = 1 - p_1$, the LP for player 1 is:

Player 1: Find p_1 to maximise V_1

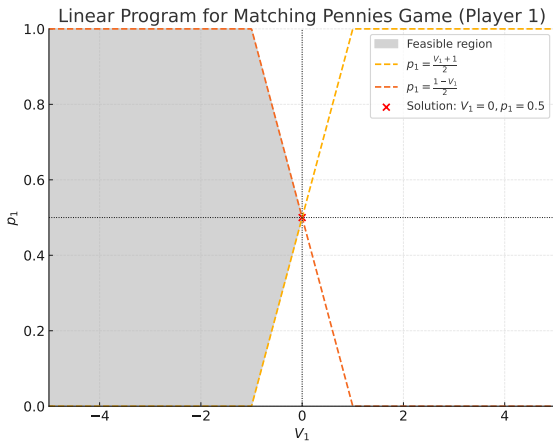
$$2p_1 - 1 \geq V_1,$$

$$\text{Subject to: } -2p_1 + 1 \geq V_1,$$

$$0 \leq p_1 \leq 1.$$

Similarly, we can use the simplex method to find the Nash equilibrium strategy for player 2.

Example: LP for Zero-Sum Games



Plot: The plot of p_1 over V_1 shows the grey area where the constraints are satisfied. The solution is:

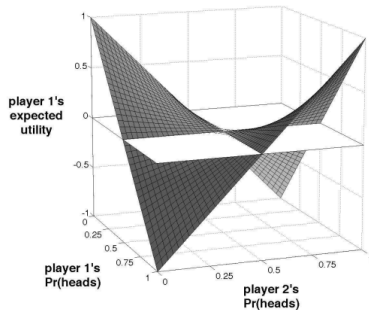
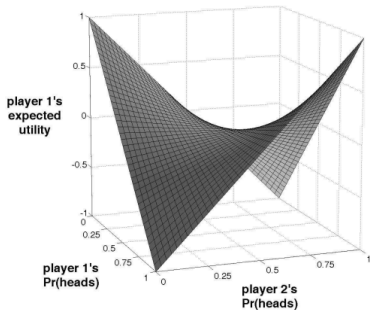
$$p_1 = 0.5, \quad V_1 = 0.$$

The Saddle Point in Matching Pennies

Definition: At a saddle point, any deviation of the agent lowers their utility and increases the utility of the other agent.

Illustration:

- The 3D plot shows player 1's expected utility versus player 2's strategy (probability of heads).
- A plane at $z = 0$ highlights the saddle point.



LP for Zero-Sum Games (in Scalar Form)

- Consider a two-player, zero-sum game $G = (\{1, 2\}, A_1 \times A_2, (u_1, u_2))$.
- U_1^* : the expected utility for player 1 in NE.
- s_i^j is the probability of player i choosing action j .
- If zero-sum, then $U_1^* = -U_2^*$ (the value of the game).

The LP for Player 1 (Primal): Maximise U_1^* subject to:

$$\begin{aligned}\sum_{j \in A_1} u_1(a_1^j, a_2^k) s_1^j &\geq U_1^*, \quad \forall k \in A_2, \\ \sum_{j \in A_1} s_1^j &= 1, \quad s_1^j \geq 0, \quad \forall j \in A_1.\end{aligned}$$

The LP for Player 2 (Dual): Minimise U_1^* subject to:

$$\begin{aligned}\sum_{k \in A_2} u_1(a_1^j, a_2^k) s_2^k &\leq U_1^*, \quad \forall j \in A_1, \\ \sum_{k \in A_2} s_2^k &= 1, \quad s_2^k \geq 0, \quad \forall k \in A_2.\end{aligned}$$

LP is typically solved using the [simplex method](#).

LP for Zero-Sum Games (in scalar form) - Note 1

Note 1: All utility terms $u_1(\cdot)$ are constants in the linear program, while the mixed strategy terms s_2^k and U_1^* are variables.

- s_i^j is the probability of player i choosing action j .
- If the game is zero-sum, then $U_1^* = -U_2^*$.

The LPs:

- Primal LP for player 1 maximises U_1^* .
- Dual LP for player 2 minimises U_1^* .

LP for Zero-Sum Games (in scalar form) - Note 2

Note 2: (Focus on Dual:) The **first constraint** states that for every pure strategy j of player 1, their expected utility for playing $j \in A_1$, given player 2's mixed strategy s_2 , is at most U_1^* .

- Pure strategies for which the expected utility is exactly U_1^* will be in player 1's **best response set**, i.e., $s_1^j > 0$.
- Pure strategies leading to lower expected utility will not.

The LP for Player 2 (Dual): Minimise U_1^* subject to:

$$\sum_{k \in A_2} u_1(a_1^j, a_2^k) s_2^k \leq U_1^*, \quad \forall j \in A_1,$$

$$\sum_{k \in A_2} s_2^k = 1, \quad s_2^k \geq 0, \quad \forall k \in A_2.$$

LP for Zero-Sum Games (in scalar form) - Note 3

Note 3: The LP will choose player 2's mixed strategy to **minimise** U_1^* subject to the constraints.

- Player 2 plays the mixed strategy that minimises the utility player 1 can gain by playing their best response.
- This is the **minimax solution**, finding player 2's optimal strategy.

The LP for Player 2 (Dual): Minimise U_1^* subject to:

$$\sum_{k \in A_2} u_1(a_1^j, a_2^k) s_2^k \leq U_1^*, \quad \forall j \in A_1,$$

$$\sum_{k \in A_2} s_2^k = 1, \quad s_2^k \geq 0, \quad \forall k \in A_2.$$

LP for Zero-Sum Games (in scalar form) - Note 4

Note 4: This LP reverses the roles of player 1 and player 2 in the constraints.

- The objective is to maximise U_1^* , as player 1 wants to maximise their own payoffs.
- This finds player 1's optimal strategy through the **maximin solution**.

The LP for Player 1 (Primal): Maximise U_1^* subject to:

$$\sum_{j \in A_1} u_1(a_1^j, a_2^k) s_1^j \geq U_1^*, \quad \forall k \in A_2,$$
$$\sum_{j \in A_1} s_1^j = 1, \quad s_1^j \geq 0, \quad \forall j \in A_1.$$

LP with Slack Variables

One can also give a formulation equivalent to our LP (for player 2) by introducing **slack variables** r_1^j for every $j \in A_1$ and replacing the inequality constraints with equality constraints.

The LP for Player 2:

Minimise U_1^* subject to:

$$\sum_{k \in A_2} u_1(a_1^j, a_2^k) \cdot s_2^k + r_1^j = U_1^*, \quad \forall j \in A_1,$$

$$\sum_{k \in A_2} s_2^k = 1, \quad s_2^k \geq 0, \quad \forall k \in A_2,$$

$$r_1^j \geq 0, \quad \forall j \in A_1.$$

Equivalence: Each slack variable must be non-negative, ensuring the constraints are satisfied.

Exercise: Odd or Even (Write Down LP solutions)⁸

Rules:

- Players I&II simultaneously call out one of the numbers 1 or 2.
- Player I wins if the sum of the numbers is **odd**.
- Player II wins if the sum of the numbers is **even**.
- The amount paid to the winner by the loser is always the sum of the numbers in dollars.

Actions:

- Action set for Player I: $X = \{1, 2\}$.
- Action set for Player II: $Y = \{1, 2\}$.

Payoff Matrix for Player I:

$A(x, y)$ = Player I's winnings = Player II's losses.

	$y = 1$	$y = 2$
$x = 1$	-2	+3
$x = 2$	+3	-4

⁸Thomas S. Ferguson. *Game Theory*. Second. UCLA, 2014.

Conclusion and Next Steps

Summary:

- We explored **zero-sum games**, their formulation, and how to compute Nash equilibria using **linear programming (LP)**.
- The **Minimax Theorem** was discussed, demonstrating its equivalence to solving an LP dual problem.
- We derived optimal strategies using **strong duality**, showing the connection between primal and dual formulations.

Next Topic: General-Sum Games and L-H Algorithms

- Moving beyond zero-sum, we will introduce **general-sum games**, where cooperation and competition coexist.
- We will study **L-H algorithms**, which computes an equilibrium strategy in general-sum settings.

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Appendix 1: Minimax Game in Standard LP Forms (1/4)⁹

Consider a 2-player zero-sum game defined by an m -by- n payoff matrix R for the row player. The column player's payoff is $-R$:

- Row player plays row i and column player plays column j .
- Ensure all entries in R are positive by adding a constant (translated values do not affect the game outcome).
- Replace $x^T \mathbf{1} = 1$ with $x^T \mathbf{1} \leq 1$, as the maximum still occurs when $x^T \mathbf{1} = 1$. the same case for y .

① Rewrite LP(1):

$$\begin{aligned} & \text{maximise } z, \\ & \text{subject to } x^T R \geq z \mathbf{1}^T, x^T \mathbf{1} \leq 1, x_i \geq 0 \quad \forall i. \end{aligned}$$

② Derive its dual form (LP(2)):

$$\begin{aligned} & \text{minimise } z', \\ & \text{subject to } -y^T R^T + z' \mathbf{1}^T \geq 0, y^T \mathbf{1} \leq 1, y_j \geq 0 \quad \forall j. \end{aligned}$$

⁹<https://www.cs.cmu.edu/~avrim/451/lectures/lect1026.pdf>

Appendix 1: Minimax Game in Standard LP Forms (2/4)

Concatenating x and z gives its standard primal form for LP(1):

$$\text{Maximise: } \mathbf{c}^T \mathbf{x} = z,$$

$$\text{Subject to: } \mathbf{x} = \begin{bmatrix} z \\ x_1 \\ \vdots \\ x_m \end{bmatrix} \geq 0, \quad \mathbf{A}\mathbf{x} \leq \mathbf{b},$$

$$\mathbf{x} = \begin{bmatrix} z \\ x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} 1 & -R^T & \cdots & \vdots \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \cdots & \cdots & \cdot \\ 0 & 1 & \cdots & 1 \end{bmatrix}.$$

Appendix 1: Minimax Game in Standard LP Forms (3/4)

Dual LP: Let $\mathbf{y}^T = (y_1, y_2, \dots, y_n, z')$. The dual problem is:

Minimise: $\mathbf{y}^T \mathbf{b} = z'$,

Subject to: $\mathbf{A}^T \mathbf{y} \geq \mathbf{c}$,

$$\text{where: } \mathbf{A}^T = \begin{bmatrix} 1 & 1 & \dots & 1 & 0 \\ -R_{11} & -R_{12} & \dots & -R_{1n} & 1 \\ -R_{21} & -R_{22} & \dots & -R_{2n} & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ -R_{m1} & -R_{m2} & \dots & -R_{mn} & 1 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

Interpretation:

- z' represents the value of the game for the row player.
- $y_1, \dots, y_n$ represents the column player's mixed strategy.
- The column player minimises w to ensure the row player's payoff is at most w regardless of the row played.

Appendix 1: Minimax Game in Standard LP Forms (4/4)

Key Observations:

- The maximum value of z in the primal problem equals the minimum value of w in the dual problem, as guaranteed by **Strong Duality**.
- This equivalence demonstrates the **Minimax Theorem**, stating:

$$\max_x \min_y x^T R y = \min_y \max_x x^T R y.$$

- The duality between these LPs directly supports the minimax relationship.

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Appendix 2: Duality in Linear Programming

We refer to section 2 in¹⁰ for an intuitive explanation of duality in linear programming.

¹⁰Princeton University COS 521 Staff. *Lecture Notes 7-8: Minimax Theorem and Zero-Sum Games*. <https://www.cs.princeton.edu/courses/archive/spring13/cos521/notes/lecture7-8.pdf>. Accessed: January 12, 2025. 2013.

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Minimax Theorem

For any finite two-player zero-sum game represented by an $m \times n$ payoff matrix A , let:

- X and Y be the mixed strategy spaces of the two players.
- The row player's worst-case payoff: $u(x) = \min_j xA_{.j}$
- The column player's best-case payoff: $w(y) = \max_i A_{i.}y^t$, where t is transpose.
- Define: $u^* = \max_{x \in X} u(x)$, $w^* = \min_{y \in Y} w(y)$

Then, the **minimax theorem** states:

$$\max_{x \in X} \min_j xA_{.j} = \min_{y \in Y} \max_i A_{i.}y^t = V. \quad (15)$$

Proof Outline¹¹

We prove the minimax theorem by **induction** on the number of rows in A .

- **Base Case:** If A is 1×1 , the result is trivially true.
- **Inductive Hypothesis:** Assume the theorem holds for all smaller matrices.
- **Inductive Step:** Show that the theorem holds for any $m \times n$ matrix A .

¹¹Guillermo Owen. "An Elementary Proof of the Minimax Theorem". In: *Management Science* 13.9 (1967), pp. 748–749. DOI: 10.1287/mnsc.13.9.748.

Step 1: Setting Up the Proof

Define:

$$a = \max_{i,j} |A_{ij}|. \quad (16)$$

Define:

$$u^* = \max_{x \in X} u(x), \quad w^* = \min_{y \in Y} w(y). \quad (17)$$

Since mixed strategy spaces are compact, u^* and w^* exist, and we always have:

$$u^* \leq w^*, \quad (18)$$

as u is the lower bound and w is upper bound of xAy^t .

Step 2: Assume Theorem is False

Suppose, for contradiction, that:

$$u^* < w^*. \quad (19)$$

Otherwise, if $u^* = w^*$, then the proof ends.

Let x and y be strategies achieving these values:

$$u(x) = u^*, \quad w(y) = w^*. \quad (20)$$

Since xAy^t is between u^* and w^* , one of the following must hold:

$$u^* < xAy^t, \quad \text{or} \quad xAy^t < w^*. \quad (21)$$

Assume:

$$xAy^t < w^*. \quad (22)$$

Then, there exists some row k such that:

$$A_k y^t = w^* - s, \quad \text{for some } s > 0. \quad (23)$$

Step 3: Construct a Smaller Matrix

Consider a smaller matrix A' , obtained by deleting row k . By the inductive hypothesis, the minimax theorem holds for A' (as it is a smaller matrix), so there exist mixed strategies $x' \in X$, $y' \in Y$ such that:

$$\min_j x'_j A_{.j} = \max_{i \neq k} A_{i.} (y')^t = v \leq a. \quad (24)$$

Step 4: Construct a New Mixed Strategy

Define new strategies as convex combinations:

$$x'' = (1 - r)x + rx', \quad (25)$$

$$y'' = (1 - r)y + ry', \quad (26)$$

where $0 < r < \frac{s}{(a-v+s)} < 1$, ensuring $x'' \in X$ and $y'' \in Y$.

Step 5: Establishing a Contradiction

By convexity of the min function¹²:

$$\min_j x'' A_{.j} \geq (1-r) \min_j x A_{.j} + r \min_j x' A_{.j}. \quad (27)$$

$$\min_j x'' A_{.j} \geq (1-r)u^* + rv. \quad \text{choose } x \text{ such as } u(x) = u^* \quad (28)$$

Similarly, by convexity of the max function:

$$\max_{i(i \neq k)} A_{i.}(y'')^t \leq (1-r)w^* + rv, \quad (29)$$

where additionally row k previously satisfied $A_k(y')^t = w^* - s$, thus also $A_k(y'')^t \leq (1-r)(w^* - s) + rv$.

¹² $\min_j [(1-\lambda)f_1(j) + \lambda f_2(j)] \geq (1-\lambda) \min_j f_1(j) + \lambda \min_j f_2(j)$. This is Jensen's inequality applied to the concave minimum function.

Step 6: Conclusions

Subtracting gives:

$$w(y'') - u(x'') \leq (1 - r)(w^* - u^*) < (w^* - u^*). \quad (30)$$

- Since we assumed $u^* < w^*$, this means that a gap exists between the row player's best worst-case payoff and the column player's worst best-case payoff.
- Constructing new mixed strategies x'' and y'' as convex combinations reduces this gap while ensuring they remain valid strategies.
- The gap between $w(y'')$ and $u(x'')$ becomes strictly smaller than $w^* - u^*$, which is impossible if u^* and w^* are the best possible values.
- This contradiction forces our assumption to be false, meaning $u^* = w^*$, proving the minimax theorem. $\square$